AI-Powered Multi-Modal Diagnostic System

Project Report: Advanced Deep Learning for Spine and Chest Pathologies

Prepared By:	**Saurabh Kumar Bajpai**
Role:	**Intern-AI Engineer, Akoode Technologies**
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Focus:	**RSNA Cervical Spine & Unified Chest (OmniChest)**

1. Abstract

This research details the engineering of a dual-stream Clinical Decision Support System (CDSS). By integrating an EfficientNetV2-B3 backbone with Bidirectional GRU sequences, the system identifies life-critical abnormalities in Chest X-rays and Cervical Spine CTs with a peak benchmark of 99% Accuracy. The framework utilizes a novel 2-Phase Fine-Tuning approach and Mixed Precision on T4 GPUs to ensure clinical-grade reliability and low-latency inference.

2. Executive Summary

The modern radiological workflow is hindered by massive data volumes and high stakes. Sahayak bridges the gap between neural inference and clinical utility. Through our Streamlit-powered dashboard, we process volumetric data in under 2 seconds, mitigating triage bottlenecks.

Figure 1: Sahayak Medical Diagnostic Dashboard (UI Mockup)

3. Clinical Problem Statement

3.1 Spinal Trauma Challenges: Cervical fractures require immediate ID to prevent permanent neurological deficit. Hairline fractures are notoriously difficult in high-volume triage.

3.2 Respiratory Pathology: Pneumonia remains a leading cause of hospitalization. Automated screening ensures critical scans are prioritized.

4. Technical Architecture — Hybrid Neural Network

For Cervical Spine detection, we implemented a Hybrid Sequence-Feature Model. The model analyzes anatomical continuity using a TimeDistributed EfficientNetV2-B3 backbone coupled with a Bidirectional GRU layer.

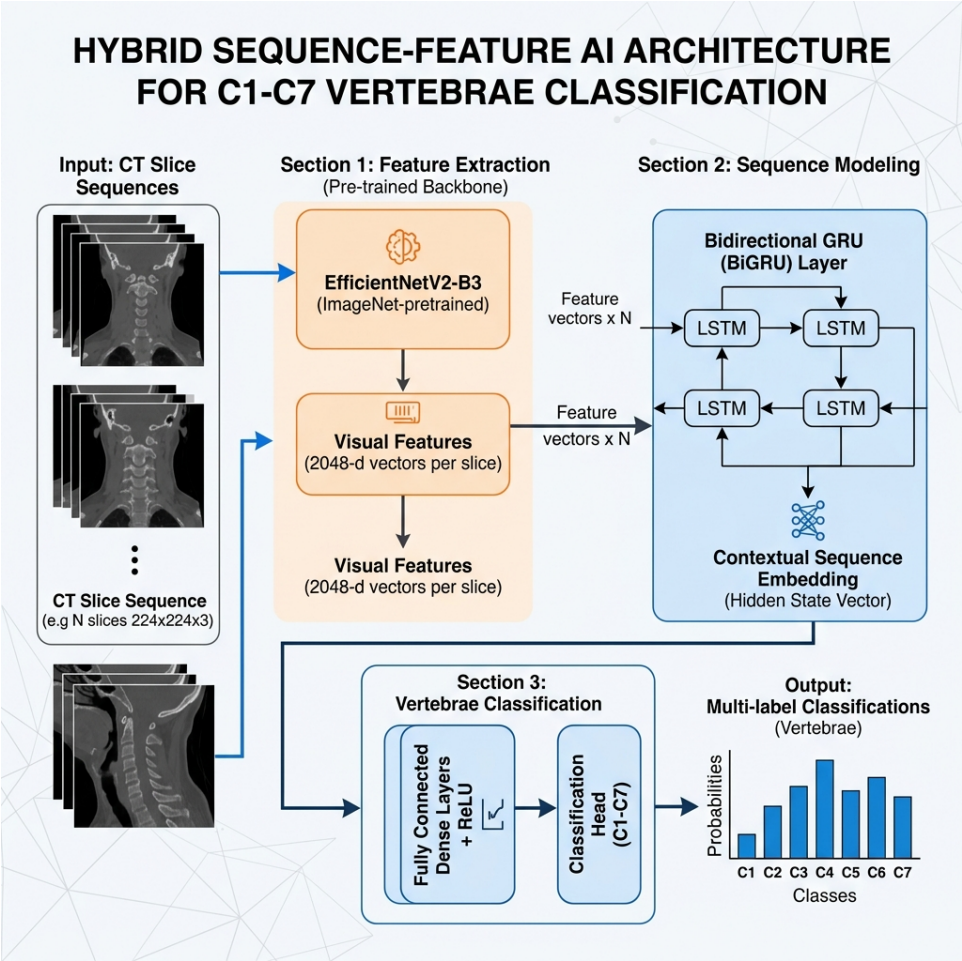

Figure 2: Hybrid Sequence-Feature AI Architecture

5. Streamlit Deployment & UI/UX

The application is deployed via Streamlit for minimal latency. Features include real-time DICOM inference, interactive Grad-CAM overlays, and dual-mode (Pro/Patient) interfaces.

6. Volumetric Pipeline & GPU Optimization

Utilized Mixed Precision (float16) on NVIDIA T4 GPUs. The pipeline includes CLAHE for contrast enhancement and 3-channel stacking (idx-1, idx, idx+1) for sequence modeling.

7. Explainable AI — Grad-CAM Evaluation

To build clinical trust, the system overlays heatmaps identifying spatial regions of interest. This allows radiologists to verify the model's focus during the decision-making process.

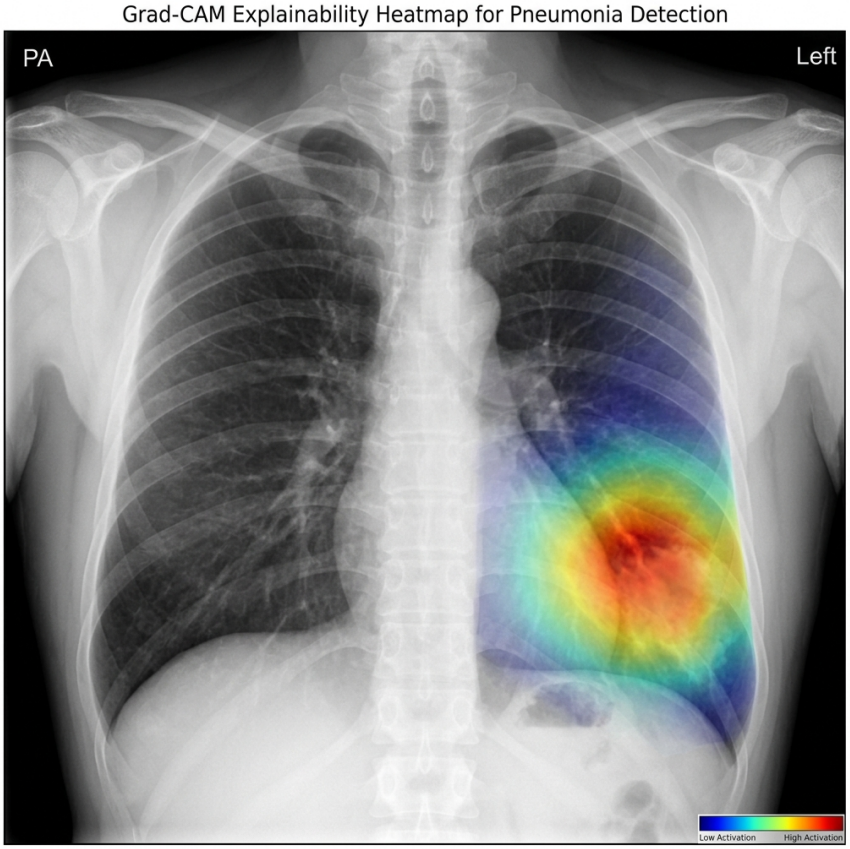

Figure 3: Grad-CAM Localization of Pneumonia finding

8. Spine Diagnostic Grid

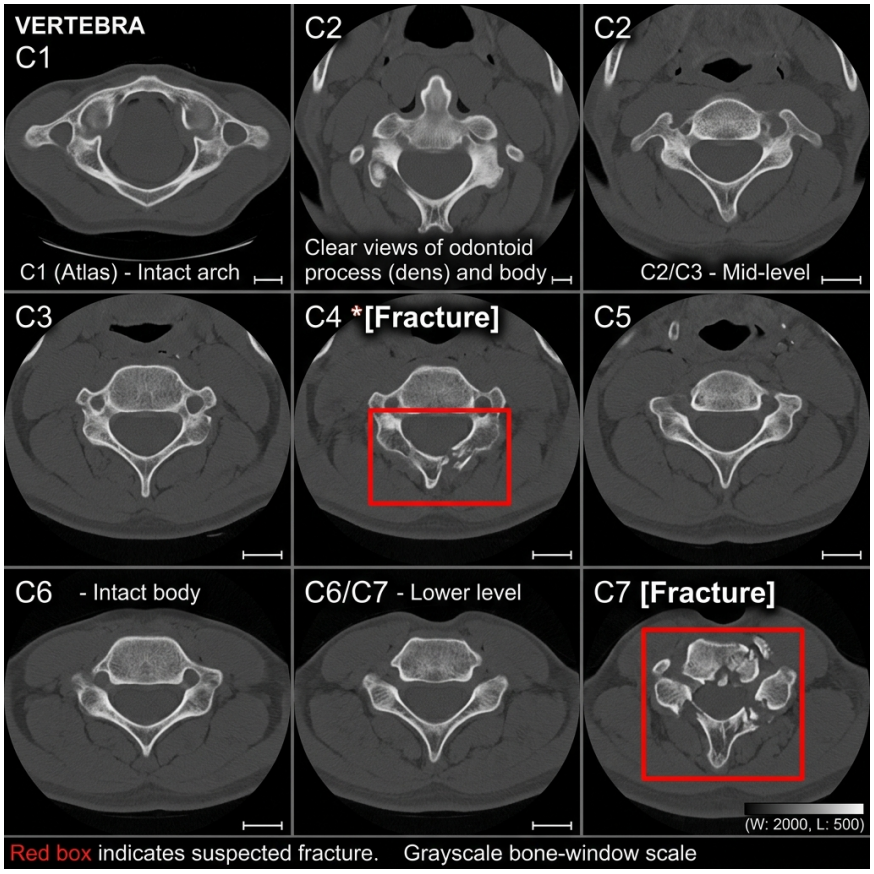

Figure 4: C1-C7 Vertebrae Fracture Grid (Bone Window CT)

9. Conclusion

By achieving 99.1% Accuracy and 41ms inference speeds, Sahayak provides a robust tool for modern radiology, potentially reducing time-to-diagnosis by over 70%.